Non-viral transfection of animal tissue culture cells

Risk assessment

- – Work with human-derived material or transgenic cell lines (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- ☐ Collect and dispose waste after inactivation as REGULATED MEDICAL WASTE

Reviewed: Apr 5, 2025

This is a bench card. Full protocol available online.

›› *Common provisions*

(1.) Prepare high-quality, endotoxin-free DNA for transfection.

(2.) One day prior to transfection, seed cells to reach 30–40% confluency on the day of transfection. Use the table below to estimate DNA and medium volumes for each format.

Reagent		**Format**	**Flasks**	**Dishes**			**Plates (per well)**			
			T-175	**15 cm**	**10 cm**	**6 cm**	**6-well**	**12-well**	**24-well**	**96-well**
		Surface area	**175 cm^2**	**145 cm^2**	**56.7 cm^2**	**21.5 cm^2**	**9.6 cm^2**	**3.5 cm^2**	**1.9 cm^2**	**1.1 cm^2**
Cells to seed		20–30%	8.2×10^6	7.0×10^6	3.0×10^6	1.1×10^6	4.2×10^5	1.8×10^5	8.4×10^4	1.4×10^4
Opti-MEM™		10 µL/cm^2	1.8 mL	1.4 mL	550 µL	200 µL	150 µL	100 µL	50 µL	25 µL
DNA	Lipofect.®	250 ng/cm^2	44.0 µg	36.0 µg	14.0 µg	5.4 µg	2.4 µg	875 ng	475 ng	275 ng
	FuGENE®	200 ng/cm^2	35.0 µg	29.0 µg	11.3 µg	4.3 µg	1.9 µg	700 ng	380 ng	220 ng
	PEI	150 ng/cm^2	26.3 µg	21.8 µg	8.5 µg	3.2 µg	1.4 µg	525 ng	285 ng	165 ng
	$CaCl_2$	450 ng/cm^2	78.8 µg	65.3 µg	25.5 µg	9.7 µg	4.3 µg	1.58 µg	855 ng	495 ng

> ***Critical:*** Adjustments may be necessary based on transfection method, cell type, or well geometry. If your format is not listed, calculate DNA and medium volume based on surface area and initial seeding density.

(3.) *Critical:* Consult the manufacturer's protocols for suspension, primary, or difficult-to-transfect cells.

A › *Transfection with Lipofectamine® 2000*

- ☐ Reduced-serum medium (Opti-MEM™)
- ☐ Lipofectamine® 2000

(1.) Dilute endotoxin-free DNA in Opti-MEM™. Use the volumes given in the planning table above.

(2.) In a separate tube, dilute 3.6 µL Lipofectamine® 2000 per microgram DNA with the same volume of Opti-MEM™ as the DNA solution. Mix gently.

> ***Critical:*** Lipofectamine® amounts should be optimized for each cell line. Adjust by increasing or decreasing 40% to balance efficiency and toxicity. Do not change the Opti-MEM™ volume.

(3.) Combine equal volumes of DNA and Lipofectamine® 2000 mixtures. Gently pipette or vortex briefly. Incubate at room temperature for 5–20 min, but not longer. 20 min

(4.) In the meantime, reduce or replace the cell culture medium volume to 50%.

(5.) Dispense the DNA–lipid complexes dropwise to each well. Gently rock to distribute; avoid swirling.

(6.) *Optional:* After 6–24 h, replace the medium.

B › *Transfection with FuGENE®*

(1.) Dilute DNA in Opti-MEM™ as above. Add FuGENE® at 3 µL per microgram DNA. Vortex briefly.

(2.) Incubate DNA–FuGENE® complexes at room temperature for 10–15 min. 15 min

(3.) Apply complexes dropwise to cells in standard culture medium. Gently rock to distribute.

C › *Transfection with polyethyleneimine (PEI)*

☐ Reduced-serum medium (Opti-MEM™)	☐ R0131 1 g/L Polyethyleneimine, 500 µL

(1.) Dilute DNA in Opti-MEM™ as above.

(2.) In a separate tube, dilute 3.0 µL polyethyleneimine (PEI) per microgram DNA with the same volume of Opti-MEM™ as the DNA solution.

(3.) Combine. Vortex briefly. Incubate DNA–PEI complexes for 10–30 min at room temperature. 10 min

(4.) Dilute 6-fold with Opti-MEM™. Apply dropwise to cells. Rock gently; avoid swirling.

(5.) *Optional:* After 6–24 h, replace the medium.

D › *Transfection with calcium chloride*

☐ R0132 2 × HEPES-buffered saline, 1 mL	☐ R0012 1 M Calcium chloride

(1.) Seed cells at 20–30% confluency already 4–6 h before transfection.

(2.) Dilute DNA in water and supplement $CaCl_2$ to a final concentration of 250 mM.

(3.) *Critical:* Drop by drop, add an equal volume of 2 × HEPES-buffered saline. Vortex after each drop. Do not let the calcium complex precipitate!

(4.) Incubate at room temperature for 15 min. 15 min

(5.) Apply dropwise to cells. Rock gently; avoid swirling.

> ***Critical:*** Calcium-mediated uptake depends on temperature and pH. Keep medium pH stable and avoid cell overgrowth.

(6.) After 6–12 h, remove the medium. Wash cells with PBS and feed fresh medium.

Recipe (available online) Troubleshooting (available online) Notes (available online)